

LAPORAN PRAKTIKUM 1

SISTEM OPERASI



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BAB I

PENDAHULUAN

1.1 LATAR BELAKANG

Sistem operasi merupakan komponen utama dalam setiap perangkat komputer yang berfungsi sebagai penghubung antara pengguna dengan perangkat keras. Tanpa sistem operasi, pengguna tidak dapat menjalankan aplikasi atau mengakses sumber daya komputer secara efisien. Linux sebagai salah satu sistem operasi open-source banyak digunakan dalam dunia akademik, industri, maupun penelitian karena kestabilan, keamanan, dan fleksibilitasnya. Melalui praktikum Sistem Operasi berbasis Linux, mahasiswa dapat memahami secara langsung konsep dasar sistem operasi seperti manajemen proses, manajemen memori, sistem berkas, dan mekanisme komunikasi antara aplikasi dengan kernel melalui system calls. Selain itu, praktikum ini juga memberikan pengalaman dalam memahami struktur internal sistem operasi, mulai dari layanan sistem (system services), pembuatan kernel module, hingga proses booting dan debugging. Pemahaman terhadap konsep dan praktik ini sangat penting karena sistem operasi merupakan fondasi utama bagi pengembangan perangkat lunak dan sistem komputer yang andal. Dengan mempelajari Linux secara praktis, mahasiswa diharapkan mampu menganalisis, memodifikasi, dan mengelola sistem operasi secara efisien, serta memiliki keterampilan teknis yang dapat diterapkan dalam lingkungan kerja profesional di bidang teknologi informasi.

1.2 RUMUSAN MASALAH

1. Bagaimana sistem operasi Linux menjalankan layanan dan mengelola sumber daya dalam proses komputasi?
2. Bagaimana peran *system calls* dan *system services* dalam interaksi antara aplikasi dan kernel sistem operasi?
3. Bagaimana cara melakukan *debugging* dan analisis performa sistem operasi menggunakan berbagai alat bantu di Linux?

1.3 TUJUAN

1. Menguasai teknik *debugging* dan analisis performa sistem untuk mendeteksi serta memperbaiki kesalahan pada program maupun sistem operasi.
2. Memahami konsep dasar dan fungsi utama sistem operasi Linux dalam pengelolaan sumber daya komputer.
3. Mempelajari cara kerja system calls, system services, serta struktur internal Linux melalui praktik langsung.

BAB II

PEMBAHASAN

2.1 Operating-System Services

Operating-System Services pada Linux merupakan sekumpulan fungsi penting yang memungkinkan sistem menjalankan dan mengelola program dengan efisien. Melalui layanan seperti eksekusi program, operasi input-output, manipulasi sistem file, komunikasi antar proses, deteksi kesalahan, serta alokasi sumber daya, sistem operasi memastikan stabilitas dan kinerja komputer tetap optimal. Perintah seperti `uname -a`, `ps aux`, `free -h`, dan `lscpu` membantu pengguna memantau kondisi sistem, sementara program seperti `os_services.c` menunjukkan bagaimana layanan OS bekerja secara langsung, misalnya dalam menulis file, membaca informasi proses, dan mengatur penggunaan memori.

2.1.1 Menjelajahi Layanan OS

- Lihat informasi sistem

```
rio_regex@DESKTOP-NVPMSHQ:~$ uname -a
Linux DESKTOP-NVPMSHQ 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
rio_regex@DESKTOP-NVPMSHQ:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.3 LTS
Release:        24.04
Codename:       noble
```

- Lihat proses yang berjalan

```
rio_regex@DESKTOP-NVPMSHQ:~$ ps aux | head -20
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.1 21668 12368 ?        Ss   20:53   0:00 /sbin/init
root         2  0.0  0.0   3072  1792 ?        Sl   20:53   0:00 /init
root         7  0.0  0.0   3072  1792 ?        Sl   20:53   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10
--log-truncate
root        41  0.0  0.2  66820 16952 ?        S<s  20:53   0:00 /usr/lib/systemd/systemd-journald
root        91  0.0  0.0  25136  6816 ?        Ss   20:53   0:00 /usr/lib/systemd/systemd-udevd
systemd+  157  0.0  0.1  21456 12416 ?        Ss   20:53   0:00 /usr/lib/systemd/systemd-resolved
systemd+  161  0.0  0.0  91024  7552 ?        Ssl  20:53   0:00 /usr/lib/systemd/systemd-timesyncd
root       169  0.0  0.0   4236  2560 ?        Ss   20:53   0:00 /usr/sbin/cron -f -P
message+  170  0.0  0.0   9624  4992 ?        Ss   20:53   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --sy
stemd-activation --syslog-only
root       177  0.0  0.1  17964  8448 ?        Ss   20:53   0:00 /usr/lib/systemd/systemd-logind
root       182  0.0  0.1 1756096 12032 ?        Ssl  20:53   0:00 /usr/libexec/wsl-pro-service -vv
syslog    197  0.0  0.0  222508  5632 ?        Ssl  20:53   0:00 /usr/sbin/rsyslogd -n -iNONE
root      201  0.0  0.0   3160  1920 hvc0    Ss+  20:53   0:00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600
vt220
root      204  0.0  0.0   3116  1664 tty1    Ss+  20:53   0:00 /sbin/agetty -o -p -- \u --noclear - linux
root      209  0.0  0.2 107032 22260 ?        Ssl  20:53   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade
-shutdown --wait-for-signal
root      301  0.0  0.0   6660  4608 pts/1    Ss   20:53   0:00 /bin/login -f
rio_reg+  344  0.0  0.1  20308 11264 ?        Ss   20:53   0:00 /usr/lib/systemd/systemd --user
rio_reg+  345  0.0  0.0  21156  3520 ?        S    20:53   0:00 (sd-pam)
rio_reg+  363  0.0  0.0   6072  4864 pts/1    S+   20:53   0:00 -bash
```

- Lihat penggunaan memori

```
rio_regex@DESKTOP-NVPMSHQ:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           7.4Gi       526Mi         6.9Gi         3.6Mi       165Mi         6.9Gi
Swap:           2.0Gi           0B         2.0Gi
```

- **Lihat informasi CPU**

```

rio_regex@DESKTOP-NVPM5HQ:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          48 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:    0-11
Vendor ID:              AuthenticAMD
Model name:             Ryzen 5 6600H with Radeon Graphics
CPU family:             25
Model:                  68
Thread(s) per core:     2
Core(s) per socket:     6
Socket(s):              1
Stepping:               1
BogoMIPS:               6587.62
Flags:                  fpu vme de pse tsc mtr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall
                        nx mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep_good nopl tsc_reliable nonstop_tsc cpuid extd_apic1
                        d tsc_known_freq pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor
                        or lahf_lm cmp_legacy svm cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw topext perfctr_core ssbd ibr
                        s ipib stlb vmmcall fsgsbase bti1 avx2 smep bmi2 ctw smp iccid rdseed adx smap clflushopt clwb sha_ni xsave
                        opt xsavec xgetbv1 xsavec clzero xsaveerptr arat npt nrip_save tsc_scale vmcb_clean flushbyasid decodeassis
                        ts pausefilter pfthreshold v_vmsave_vmload umip vaes vpclmulqdq rdpid fsrm

Virtualization features:
  Virtualization:       AMD-V
  Hypervisor vendor:    Microsoft
  Virtualization type:  full
Caches (sum of all):
  L1d:                  192 KiB (6 instances)
  L1i:                  192 KiB (6 instances)
  L2:                   3 MiB (6 instances)
  L3:                   16 MiB (1 instance)
NUMA:
  NUMA node(s):         1
  NUMA node0 CPU(s):    0-11
Vulnerabilities:
  Gather data sampling:  Not affected
  Itlb multihit:         Not affected
  L1tf:                  Not affected
  Rds:                   Not affected
  Meltdown:              Not affected
  Mmio stale data:       Not affected
  Reg file data sampling: Not affected
  Retbleed:              Not affected
  Spec rstack overflow:  Vulnerable: Safe RET, no microcode
  Spec store bypass:     Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:            Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:            Mitigation; Retpolines; IBPB conditional; IBRS_FW; STIBP always-on; RSB filling; PBRSB-eIBRS Not affected;
                        BHI Not affected
  Srbds:                 Not affected
  Tsx async abort:       Not affected

```

- **Lihat perangkat yang terpasang**

```

rio_regex@DESKTOP-NVPM5HQ:~$ lsblk
NAME                                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda      8:0    0 388.4M  1 disk
sdb      8:16   0   186M  1 disk
sdc      8:32   0     2G  0 disk [SWAP]
sdd      8:48   0     1T  0 disk /mnt/wslg/distro
/

```

2.1.2 Program C - Menggunakan Layanan OS

```

rio_regex@DESKTOP-NVPM5HQ:~$ nano os_services.c
rio_regex@DESKTOP-NVPM5HQ:~$ gcc -o os_services os_services.c
rio_regex@DESKTOP-NVPM5HQ:~$ ./os_services
=== Program Execution Service ===
Process ID (PID): 610
Parent Process ID (PPID): 534

=== I/O Operations Service ===
File created and written successfully

=== File System Manipulation ===
File size: 23 bytes
File permissions: 644

=== Communications ===
Hostname: (null)

=== Error Detection ===
Error opening file: No such file or directory

=== Resource Allocation ===
Memory allocated successfully

```

2.2. User and Operating-System Interface

User and Operating-System Interface pada Linux adalah jembatan interaksi antara pengguna dan sistem operasi melalui Command Line Interface (CLI). Pengguna dapat menavigasi sistem menggunakan perintah seperti `pwd`, `ls`, `cd`, dan `mkdir`, serta mengelola data dengan pipelining dan redirection. Selain itu, Linux juga mendukung proses berjalan di latar belakang menggunakan `&`, `jobs`, dan `fg`. Dalam praktik program `simple_shell.c`, konsep ini ditunjukkan melalui pembuatan shell sederhana yang menggunakan `fork()` dan `execvp()`.

untuk mengeksekusi perintah pengguna, sehingga menggambarkan bagaimana sistem operasi menangani proses dan komunikasi antar program secara langsung.

2.2.1 Command Line Interface (CLI)

- Navigasi dasar

```
rio_regex@DESKTOP-NVPM5HQ:~$ pwd
/home/rio_regex
rio_regex@DESKTOP-NVPM5HQ:~$ ls -la
total 268
drwxr-xr-x 6 rio_regex rio_regex 4096 Oct 13 22:08 .
drwxr-xr-x 3 root root 4096 Oct 7 22:00 ..
-rw-r--r-- 1 rio_regex rio_regex 8560 Oct 13 21:34 .bash_history
-rw-r--r-- 1 rio_regex rio_regex 220 Oct 7 22:00 .bash_logout
-rw-r--r-- 1 rio_regex rio_regex 3771 Oct 7 22:00 .bashrc
drwx----- 2 rio_regex rio_regex 4096 Oct 7 22:01 .cache
drwx----- 3 rio_regex rio_regex 4096 Oct 9 18:31 .config
-rw----- 1 rio_regex rio_regex 20 Oct 9 19:32 .lesshst
drwxr-xr-x 3 rio_regex rio_regex 4096 Oct 7 22:13 .local
-rw-rw-r-- 1 rio_regex rio_regex 0 Oct 13 08:10 .motd_shown
-rw-r--r-- 1 rio_regex rio_regex 807 Oct 7 22:00 .profile
-rw-r--r-- 1 rio_regex rio_regex 0 Oct 7 23:20 .sudo_as_admin_successful
-rw-r--r-- 1 rio_regex rio_regex 404 Oct 13 09:01 fifo_reader.c
-rw-r--r-- 1 rio_regex rio_regex 513 Oct 13 09:00 fifo_writer.c
-rw-r--r-- 1 rio_regex rio_regex 5208 Oct 11 14:35 file_monitor.c
-rw-r--r-- 1 rio_regex rio_regex 615 Oct 13 08:51 fork_demo.c
-rwxr-xr-x 1 rio_regex rio_regex 16680 Oct 13 09:25 ipc_benchmark
-rw-r--r-- 1 rio_regex rio_regex 1797 Oct 13 09:24 ipc_benchmark.c
-rw-r--r-- 1 rio_regex rio_regex 703 Oct 13 08:54 multiple_fork.c
-rwxr-xr-x 1 rio_regex rio_regex 16488 Oct 13 22:08 os_services
-rw-r--r-- 1 rio_regex rio_regex 1315 Oct 13 22:07 os_services.c
-rw-r--r-- 1 rio_regex rio_regex 767 Oct 13 08:56 pipe_demo.c
-rwxr-xr-x 1 rio_regex rio_regex 16096 Oct 13 08:47 process_basic
-rw-r--r-- 1 rio_regex rio_regex 216 Oct 13 08:47 process_basic.c
-rwxr-xr-x 1 rio_regex rio_regex 22208 Oct 10 11:24 process_manager
-rw-r--r-- 1 rio_regex rio_regex 14749 Oct 10 11:23 process_manager.c
-rw-r--r-- 1 rio_regex rio_regex 597 Oct 13 08:53 process_wait.c
drwxr-xr-x 2 rio_regex rio_regex 4096 Oct 9 13:58 projects
-rwxr-xr-x 1 rio_regex rio_regex 16224 Oct 13 09:02 reader
-rwxr-xr-x 1 rio_regex rio_regex 16176 Oct 13 09:12 shm_reader
-rw-r--r-- 1 rio_regex rio_regex 576 Oct 13 09:11 shm_reader.c
-rwxr-xr-x 1 rio_regex rio_regex 16352 Oct 13 09:12 shm_writer
-rw-r--r-- 1 rio_regex rio_regex 689 Oct 13 09:10 shm_writer.c
-rw-r--r-- 1 rio_regex rio_regex 0 Oct 13 09:11 shmfile
-rw-r--r-- 1 rio_regex rio_regex 23 Oct 13 22:08 test_file.txt
-rwxr-xr-x 1 rio_regex rio_regex 16352 Oct 13 09:07 writer
rio_regex@DESKTOP-NVPM5HQ:~$ cd /tmp
rio_regex@DESKTOP-NVPM5HQ:~$ tmp$ mkdir test_dir
rio_regex@DESKTOP-NVPM5HQ:~$ tmp$ cd test_dir
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$
```

- Pipelining

```
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$ ls -l | grep "^d"
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$ ps aux | grep bash | wc -l
3
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$
```

- Redirection

```
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$ echo "Hello World" > output.txt
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$ cat output.txt
Hello World
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$ cat output.txt >> output.txt
cat: output.txt: input file is output file
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$ cat < output.txt
Hello World
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$
```

- Background processes

```
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$ sleep 30 &
[1] 715
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$ jobs
[1]+  Running                  sleep 30 &
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$ fg %1
sleep 30
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$
```

2.2.2 Program C - Shell Sederhana

```
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$ nano simple_shell.c
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$ gcc -o simple_shell simple_shell.c
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$ ./simple_shell
myshell> echo HELLO RIO
HELLO RIO
myshell> ^Z
[1]+  Stopped                  ./simple_shell
rio_regex@DESKTOP-NVPM5HQ:~/tmp/test_dir$
```

2.3 System Calls

System Calls merupakan mekanisme penting yang memungkinkan program aplikasi berinteraksi langsung dengan kernel sistem operasi. Melalui system call, program dapat meminta layanan seperti pembuatan proses, manajemen file, maupun pemeliharaan informasi sistem. Dalam praktik 3.1, **Process Control System Calls** seperti `fork()`, `exec()`, dan `wait()` digunakan untuk membuat proses baru, mengeksekusi program lain, serta menunggu proses anak selesai. Pada praktik 3.2, **File Management System Calls** seperti `open()`, `write()`, `read()`, `close()`, dan `unlink()` menunjukkan bagaimana program dapat membuat, menulis, membaca, menutup, dan menghapus file secara langsung melalui kernel. Sementara itu, praktik 3.3 menggunakan **Information Maintenance System Calls** seperti `getpid()`, `getuid()`, `time()`, dan `times()` untuk menampilkan informasi proses, pengguna, waktu sistem, serta aktivitas CPU. Dengan demikian, system calls berperan penting dalam menjembatani komunikasi antara aplikasi pengguna dan sumber daya inti sistem operasi Linux.

2.3.1 Process Control System Calls

```
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ nano process_syscalls.c
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ gcc -o process_syscalls process_syscalls.c
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ ./process_syscalls
=== Process Control System Calls ===
Parent PID: 783
Parent waiting for child (PID: 784)...
Child PID: 784
Child's Parent PID: 783
Child executing 'ls' command...
total 48
-rw-r--r-- 1 rio_regex rio_regex 12 Oct 13 22:38 output.txt
-rwxr-xr-x 1 rio_regex rio_regex 16360 Oct 13 23:28 process_syscalls
-rw-r--r-- 1 rio_regex rio_regex 831 Oct 13 23:26 process_syscalls.c
-rwxr-xr-x 1 rio_regex rio_regex 16528 Oct 13 23:07 simple_shell
-rw-r--r-- 1 rio_regex rio_regex 896 Oct 13 23:06 simple_shell.c
Child exited with status: 0
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$
```

2.3.2 File Management System Calls

```
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ nano file_syscalls.c
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ gcc -o file_syscalls file_syscalls.c
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ ./file_syscalls
=== File Management System Calls ===
File descriptor: 3
Bytes written: 25
File closed
Read from file: Hello from system calls!
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$
```

2.3.3 Information Maintenance System Calls

```
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ nano info_syscalls.c
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ gcc -o info_syscalls info_syscalls.c
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ ./info_syscalls
=== Information Maintenance System Calls ===
Process ID: 810
Parent Process ID: 534
User ID: 1000
Group ID: 1000
Current time: Mon Oct 13 23:40:33 2025
Clock ticks since boot: 1718937835
Sleeping for 2 seconds...
Awake!
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$
```

2.4 System Services

System Services di Linux merupakan komponen penting yang berjalan di latar belakang untuk memastikan sistem dan aplikasi berfungsi dengan baik. Layanan ini dikelola oleh systemd, yaitu sistem init modern yang bertanggung jawab untuk mengatur proses booting, manajemen service, dan kontrol proses daemon. Dalam praktik 4.1, pengguna dapat melihat semua layanan aktif dengan perintah `systemctl list-units --type=service`, memeriksa status layanan seperti ssh menggunakan `systemctl status ssh`, serta meninjau log aktivitas melalui

journalctl -u ssh. Selain itu, systemctl list-unit-files --type=service --state=enabled digunakan untuk mengetahui service yang otomatis dijalankan saat boot.

Pada praktik 4.2, pengguna mempelajari cara membuat custom system service dengan program daemon sederhana bernama *my_daemon.c*. Program ini dijalankan di latar belakang menggunakan mekanisme fork(), setsid(), dan syslog() untuk mencatat aktivitas ke log sistem. File konfigurasi service kemudian dibuat di /etc/systemd/system/my_daemon.service, yang menentukan bagaimana daemon dijalankan dan dikelola oleh systemd. Dengan menggunakan perintah systemctl, pengguna dapat memulai, menghentikan, memantau, dan mengatur agar service berjalan otomatis saat startup. Melalui praktik ini, dipahami bahwa systemd dan system services berperan penting dalam menjaga stabilitas, otomatisasi, dan pengelolaan proses di sistem operasi Linux secara efisien dan terstruktur.

2.4.1 Menggunakan system

- **Lihat semua service**

```
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ systemctl list-units --type=service
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
console-getty.service              loaded active running Console Getty
console-setup.service              loaded active exited Set console font and keymap
cron.service                        loaded active running Regular background program processing daemon
dbus.service                       loaded active running D-Bus System Message Bus
getty@tty1.service                 loaded active running Getty on tty1
keyboard-setup.service             loaded active exited Set the console keyboard layout
kmod-static-nodes.service          loaded active exited Create List of Static Device Nodes
rsyslog.service                   loaded active running System Logging Service
setvtrgb.service                  loaded active exited Set console scheme
snapd.seeded.service              loaded active exited Wait until snapd is fully seeded
systemd-binfmt.service            loaded active exited Set Up Additional Binary Formats
systemd-journal-flush.service      loaded active exited Flush Journal to Persistent Storage
systemd-journald.service           loaded active running Journal Service
systemd-logind.service             loaded active running User Login Management
systemd-modules-load.service       loaded active exited Load Kernel Modules
systemd-remount-fs.service         loaded active exited Remount Root and Kernel File Systems
systemd-resolved.service           loaded active running Network Name Resolution
systemd-sysctl.service            loaded active exited Apply Kernel Variables
systemd-timesyncd.service          loaded active running Network Time Synchronization
systemd-tmpfiles-setup-dev-early.service loaded active exited Create Static Device Nodes in /dev gracefully
systemd-tmpfiles-setup-dev.service loaded active exited Create Static Device Nodes in /dev
systemd-tmpfiles-setup.service     loaded active exited Create Volatile Files and Directories
systemd-udev-trigger.service       loaded active exited Coldplug All udev Devices
systemd-udevd.service              loaded active running Rule-based Manager for Device Events and Files
systemd-update-utmp.service        loaded active exited Record System Boot/Shutdown in UTMP
systemd-user-sessions.service      loaded active exited Permit User Sessions
unattended-upgrades.service        loaded active running Unattended Upgrades Shutdown
user-runtime-dir@1000.service      loaded active exited User Runtime Directory /run/user/1000
user@1000.service                 loaded active running User Manager for UID 1000
wsl-pro.service                   loaded active running Bridge to Ubuntu Pro agent on Windows

Legend: LOAD    + Reflects whether the unit definition was properly loaded.
          ACTIVE + The high-level unit activation state, i.e. generalization of SUB.
          SUB    + The low-level unit activation state, values depend on unit type.

30 loaded units listed. Pass --all to see loaded but inactive units, too.
To show all installed unit files use 'systemctl list-unit-files'.
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$
```

- **Status service**

```
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
 TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$
```

- **Lihat log service**

```
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ journalctl -u ssh -n 20
-- No entries --
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$
```

- Lihat service yang berjalan saat boot

```

rio_rege@DESKTOP-NVPM5HQ:/tmp/test_dir$ systemctl list-unit-files --type=service --state=enabled
UNIT FILE                                STATE PRESET
apparmor.service                        enabled enabled
apport.service                          enabled enabled
cloud-config.service                   enabled enabled
cloud-final.service                     enabled enabled
cloud-init-local.service                 enabled enabled
cloud-init.service                      enabled enabled
console-setup.service                   enabled enabled
cron.service                            enabled enabled
dmesg.service                           enabled enabled
e2scrub_reap.service                    enabled enabled
getty@.service                           enabled enabled
keyboard-setup.service                  enabled enabled
landscape-client.service                 enabled enabled
networkd-dispatcher.service             enabled enabled
rsyslog.service                         enabled enabled
setvtrgb.service                        enabled enabled
snapd.apparmor.service                  enabled enabled
snapd.autoimport.service                enabled enabled
snapd.core-fixup.service                 enabled enabled
snapd.recovery-chooser-trigger.service   enabled enabled
snapd.seeded.service                    enabled enabled
snapd.service                           enabled enabled
snapd.system-shutdown.service            enabled enabled
systemd-pstore.service                  enabled enabled
systemd-resolved.service                 enabled enabled
systemd-timesyncd.service                enabled enabled
ua-reboot-cmds.service                   enabled enabled
ubuntu-advantage.service                 enabled enabled
unattended-upgrades.service              enabled enabled
wsl-pro.service                          enabled enabled

30 unit files listed.
rio_rege@DESKTOP-NVPM5HQ:/tmp/test_dir$

```

2.4.2 Membuat Custom System Service

```

rio_rege@DESKTOP-NVPM5HQ:/tmp/test_dir$ nano my_daemon.c
rio_rege@DESKTOP-NVPM5HQ:/tmp/test_dir$ gcc -o my_daemon my_daemon.c
rio_rege@DESKTOP-NVPM5HQ:/tmp/test_dir$

```

```

rio_rege@DESKTOP-NVPM5HQ:/tmp/test_dir$ nano my_daemon.c
rio_rege@DESKTOP-NVPM5HQ:/tmp/test_dir$ gcc -o my_daemon my_daemon.c
rio_rege@DESKTOP-NVPM5HQ:/tmp/test_dir$ sudo nano /etc/systemd/system/my_daemon.service:
[sudo] password for rio_rege:
rio_rege@DESKTOP-NVPM5HQ:/tmp/test_dir$ sudo systemctl daemon-reload
rio_rege@DESKTOP-NVPM5HQ:/tmp/test_dir$ sudo systemctl start my_daemon
rio_rege@DESKTOP-NVPM5HQ:/tmp/test_dir$ sudo systemctl status my_daemon
# my_daemon.service - My Custom Daemon
Loaded: loaded (/etc/systemd/system/my_daemon.service; disabled; preset: enabled)
Active: active (running) since Tue 2025-10-14 00:14:49 WIB; 15s ago
Process: 982 ExecStart=/tmp/test_dir/my_daemon (code=exited, status=0/SUCCESS)
Main PID: 983 (my_daemon)
Tasks: 1 (limit: 9962)
Memory: 204.0K (peak: 768.0K)
CPU: 3ms
CGroup: /system.slice/my_daemon.service
└─983 /tmp/test_dir/my_daemon

Oct 14 00:14:49 DESKTOP-NVPM5HQ systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 14 00:14:49 DESKTOP-NVPM5HQ my_daemon[983]: My daemon started
Oct 14 00:14:49 DESKTOP-NVPM5HQ my_daemon[983]: Daemon is running...
Oct 14 00:14:49 DESKTOP-NVPM5HQ systemd[1]: Started my_daemon.service - My Custom Daemon.
rio_rege@DESKTOP-NVPM5HQ:/tmp/test_dir$ sudo journalctl -u my_daemon
Oct 09 16:07:50 DESKTOP-NVPM5HQ systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 09 16:07:50 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 09 16:07:50 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 16:07:50 DESKTOP-NVPM5HQ systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 16:07:50 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 1.
Oct 09 16:07:50 DESKTOP-NVPM5HQ systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 09 16:07:50 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 09 16:07:50 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 16:07:50 DESKTOP-NVPM5HQ systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 16:07:50 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 2.
Oct 09 16:07:50 DESKTOP-NVPM5HQ systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 09 16:07:50 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 09 16:07:50 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 16:07:50 DESKTOP-NVPM5HQ systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 16:07:51 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 3.
Oct 09 16:07:51 DESKTOP-NVPM5HQ systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 09 16:07:51 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 09 16:07:51 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 16:07:51 DESKTOP-NVPM5HQ systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 16:07:51 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 4.
Oct 09 16:07:51 DESKTOP-NVPM5HQ systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 09 16:07:51 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 09 16:07:51 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 16:07:51 DESKTOP-NVPM5HQ systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 16:07:51 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 5.
Oct 09 16:07:51 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Start request repeated too quickly.
Oct 09 16:07:51 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 16:07:51 DESKTOP-NVPM5HQ systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 16:14:45 DESKTOP-NVPM5HQ systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 09 16:14:45 DESKTOP-NVPM5HQ my_daemon[1570]: My daemon is running.
Oct 09 16:15:29 DESKTOP-NVPM5HQ systemd[1]: my_daemon.service: Deactivated successfully.
Oct 09 16:15:29 DESKTOP-NVPM5HQ systemd[1]: Stopped my_daemon.service - My Custom Daemon.
-- Next: 297f4a8b3a39e46888c1f46470126e --
Oct 14 00:14:49 DESKTOP-NVPM5HQ systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 14 00:14:49 DESKTOP-NVPM5HQ my_daemon[983]: My daemon started
Oct 14 00:14:49 DESKTOP-NVPM5HQ my_daemon[983]: Daemon is running...
Oct 14 00:14:49 DESKTOP-NVPM5HQ systemd[1]: Started my_daemon.service - My Custom Daemon.
Oct 14 00:15:21 DESKTOP-NVPM5HQ my_daemon[983]: Daemon is running...

```

2.5 Linkers and Loaders

Dalam sistem operasi Linux, *linkers* dan *loaders* memiliki peran penting dalam proses pembuatan dan eksekusi program. Linker bertugas menggabungkan berbagai *object file* hasil kompilasi (seperti main.o dan math_ops.o) menjadi satu *executable file* yang utuh, misalnya program. Pada proses ini, linker menyelesaikan referensi antar file, seperti pemanggilan fungsi add() dan multiply() yang didefinisikan di file lain. Loader, di sisi lain, bertugas memuat program tersebut ke dalam memori saat dijalankan, menyiapkan ruang memori, serta menghubungkan dependensi *shared library* bila menggunakan *dynamic linking*.

Dalam praktik 5.1, proses linking dilakukan setelah tahap preprocessing, compiling, dan assembling menggunakan perintah gcc main.o math_ops.o -o program. Hasilnya dapat dianalisis dengan nm untuk melihat simbol, dan ldd untuk melihat dependensi. Melalui praktik ini, mahasiswa memahami perbedaan antara *static* dan *dynamic linking*, serta bagaimana membuat *shared library* (libmathops.so) yang dapat digunakan berulang kali tanpa perlu dikompilasi ulang.

2.5.1 Proses Linking

- **Buat file math_ops.h**

```
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ nano math_ops.h
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ |
```

- **Buat file math_ops.c**

```
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ nano math_ops.c
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ |
```

- **Buat main .c**

```
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ nano main.c
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ |
```

- **Preprocessing**

```
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ gcc -E main.c -o main.i
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ gcc -E math_ops.c -o math_ops.i
```

- **Compilation (assembly)**

```
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ gcc -S main.c -o main.s
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ gcc -S math_ops.c -o math_ops.s
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ |
```

- **Assembly (object files)**

```
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ gcc -c main.c -o main.o
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ gcc -c math_ops.c -o math_ops.o
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ |
```

- **Linking**

```
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ gcc main.o math_ops.o -o program
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ |
```

- Lihat symbols dalam object file

```

rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ nm main.o
0000000000000000 T add
0000000000000000 T main
0000000000000000 U multiply
0000000000000000 U printf
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ nm math_ops.o
0000000000000000 T add
0000000000000018 T multiply
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ nm program
0000000000003dc8 d _DYNAMIC
0000000000003fb8 d _GLOBAL_OFFSET_TABLE_
0000000000002000 R _IO_stdin_used
0000000000002148 w _ITM_deregisterTMCloneTable
0000000000002148 w _ITM_registerTMCloneTable
0000000000002148 r _FRAME_END_
000000000000201c r __GNU_EH_FRAME_HDR
0000000000004010 D __TMC_END__
000000000000038c r __abi_tag
0000000000004010 B __bss_start
0000000000004000 w __cxa_finalize@GLIBC_2.2.5
0000000000004000 D __data_start
0000000000001100 t __do_global_dtors_aux
0000000000003dc0 d __do_global_dtors_aux_fini_array_entry
0000000000004000 D __dso_handle
0000000000003db8 d __frame_dummy_init_array_entry
0000000000004000 w __gmon_start__
0000000000004010 U __libc_start_main@GLIBC_2.34
0000000000004010 D _edata
0000000000004018 B _end
00000000000011d4 T _fini
0000000000001000 T _init
0000000000001000 T _start
00000000000011a2 T add
0000000000004010 b completed.0
0000000000004000 W data_start
0000000000001090 t deregister_tm_clones
0000000000001140 t frame_dummy
0000000000001149 T main
00000000000011ba T multiply
00000000000010c0 t register_tm_clones
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$

```

- Lihat dependencies

```

rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ ldd program
linux-vdso.so.1 (0x00007ffec454a000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x0000780da5000000)
/lib64/ld-linux-x86-64.so.2 (0x0000780da529a000)
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$

```

2.5.2 Static vs Dynamic Linking

- Static linking

```

rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ gcc main.c math_ops.c -o program_static -static
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$

```

- Dynamic linking (default)

```

rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ gcc main.c math_ops.c -o program_dynamic
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$

```

- Bandingkan ukuran

```

rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ ls -lh program_static program_dynamic
-rwxr-xr-x 1 rio_regex rio_regex 16K Oct 14 00:58 program_dynamic
-rwxr-xr-x 1 rio_regex rio_regex 767K Oct 14 00:57 program_static
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$

```

- Lihat dependencies

```

rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ ldd program_static
not a dynamic executable
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ ldd program_dynamic
linux-vdso.so.1 (0x00007ffda4199000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007ce8cee00000)
/lib64/ld-linux-x86-64.so.2 (0x00007ce8cflc8000)
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$

```

2.5.3 Membuat dan Menggunakan Shared Library

- Compile sebagai shared library

```

rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ gcc -c -fPIC math_ops.c -o math_ops.o
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ gcc -shared -o libmathops.so math_ops.o
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$

```

- Compile program yang menggunakan library

```

rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ gcc main.c -L. -lmathops -o program_shared
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$

```

- Jalankan dengan LD_LIBRARY_PATH

```
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ LD_LIBRARY_PATH=./program_shared
5 + 3 = 8
5 * 3 = 15
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$
```

- Lihat informasi loader

```
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ readelf -d program_shared
Dynamic section at offset 0x2da8 contains 28 entries:
  Tag                Type              Name/Value
0x0000000000000001 (NEEDED)           Shared library: [libmathops.so]
0x0000000000000001 (NEEDED)           Shared library: [libc.so.6]
0x000000000000000c (INIT)             0x1000
0x000000000000000c (INIT)             0x11e4
0x0000000000000019 (INIT_ARRAY)         0x3d08
0x000000000000001b (INIT_ARRAYSZ)      8 (bytes)
0x000000000000001a (FINI_ARRAY)         0x3da0
0x000000000000001c (FINI_ARRAYSZ)      8 (bytes)
0x000000006ffffef5 (GNU_HASH)          0x3b0
0x0000000000000005 (STRTAB)            0x4b0
0x0000000000000006 (SYMTAB)            0x3d8
0x000000000000000a (STRSZ)             170 (bytes)
0x000000000000000b (SYMINFO)           24 (bytes)
0x0000000000000015 (DEBUG)             0x0
0x0000000000000003 (PLTGOT)            0x3fa8
0x0000000000000002 (PLTRELSZ)          72 (bytes)
0x0000000000000014 (PLTREL)            RELA
0x0000000000000017 (JMPREL)            0x660
0x0000000000000007 (RELA)              0x5a0
0x0000000000000008 (RELASZ)            192 (bytes)
0x0000000000000009 (RELAENT)           24 (bytes)
0x000000000000001e (FLAGS)             BIND_NOW
0x000000006fffffb (FLAGS_1)           Flags: NOW PIE
0x000000006fffffe (VERNEED)            0x570
0x000000006fffff0 (VERNEEDNUM)         1
0x000000006fffff0 (VERSYM)            0x55a
0x000000006fffff9 (RELACOUNT)          3
0x0000000000000000 (NULL)             0x0
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$
```

2.6 Why Applications Are Operating-System Specific

Aplikasi bersifat *operating-system specific* karena setiap sistem operasi memiliki arsitektur internal, format biner, serta sistem pemanggilan (*system calls*) yang berbeda. *System call* adalah antarmuka utama antara aplikasi dan kernel, yang mengatur komunikasi seperti manajemen proses, file, dan perangkat keras. Misalnya, pada Linux fungsi `getpid()` digunakan untuk mendapatkan ID proses melalui pustaka `<unistd.h>`, sedangkan di Windows fungsi serupa membutuhkan pustaka dan API berbeda seperti `<windows.h>`.

Pada Praktik 6.1, file `os_specific.c` menunjukkan perbedaan ini dengan menggunakan *directive preprocessor* (`#ifdef _linux_`, `#elif _WIN32`) untuk menyesuaikan kode berdasarkan sistem operasi yang digunakan. Program tersebut mencetak informasi OS dengan fungsi `uname()` pada Linux atau menggunakan API Windows ketika dijalankan di sistem Microsoft.

Selanjutnya, Praktik 6.2 menganalisis *binary format* hasil kompilasi menggunakan perintah seperti `file`, `readelf`, dan `objdump`. Di Linux, executable berformat ELF (Executable and Linkable Format), sedangkan Windows menggunakan PE (Portable Executable). Perbedaan format ini mempertegas bahwa program yang dikompilasi di satu sistem operasi tidak dapat langsung dijalankan di OS lain tanpa *recompilation* atau *compatibility layer* seperti Wine.

2.6.1 System Call Differences

```
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ nano os_specific.c
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$
```

2.6.2 Binary Format Analysis

- Lihat format executable

```
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ file program
program: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=0c5145e0b63b470
95af229368edc342f23076fc2, for GNU/Linux 3.2.0, not stripped
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$
```

- **Lihat header ELF**

```
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ readelf -h program
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00 00
  Class:       ELF64
  Data:       2's complement, little endian
  Version:    1 (current)
  OS/ABI:     UNIX - System V
  ABI Version: 0
  Type:       DYN (Position-Independent Executable file)
  Machine:    Advanced Micro Devices X86-64
  Version:    0x1
  Entry point address: 0x1060
  Start of program headers: 64 (bytes into file)
  Start of section headers: 14872 (bytes into file)
  Flags:      0x0
  Size of this header: 64 (bytes)
  Size of program headers: 56 (bytes)
  Number of program headers: 13
  Size of section headers: 64 (bytes)
  Number of section headers: 31
  Section header string table index: 30
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$
```

- **Lihat section headers**

```
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ readelf -S program
There are 31 section headers, starting at offset 0x1060

Section Headers:
 [Nr] Name                Type              Address            Offset
     Size               Info              File Linker       Info
-----
 [ 0]                     NULL              0000000000000000   00000000
 [ 1] .interp               PROGBITS          0000000000000318   00000318
 [ 2] .note.gnu.property    NOTE              0000000000000318   00000318
 [ 3] .note.gnu.build-id     NOTE              0000000000000368   00000368
 [ 4] .note.ABI-tag          NOTE              00000000000003c0   000003c0
 [ 5] .gnu.hash              GNU_HASH           00000000000003d8   000003d8
 [ 6] .dynsym                DYNAMIC            00000000000003e8   000003e8
 [ 7] .dynamic                DYNAMIC            00000000000003f8   000003f8
 [ 8] .gnu.version           VERSYM             0000000000000410   00000410
 [ 9] .gnu.version_r          VERNEED            0000000000000428   00000428
[10] .rela.dyn               RELA               0000000000000438   00000438
[11] .rela.plt               RELA               00000000000004c0   000004c0
[12] .init                   PROGBITS          0000000000001000   00001000
[13] .plt                    PROGBITS          0000000000001028   00001028
[14] .plt.got                PROGBITS          0000000000001040   00001040
[15] .plt.sec                PROGBITS          0000000000001058   00001058
[16] .text                   PROGBITS          0000000000001060   00001060
[17] .fini                   PROGBITS          0000000000001140   00001140
[18] .rodata                 PROGBITS          0000000000002000   00002000
[19] .eh_frame_hdr           PROGBITS          000000000000201c   0000201c
[20] .eh_frame               PROGBITS          0000000000002000   00002000
[21] .init_array              INIT_ARRAY        0000000000003db8   00003db8
[22] .fini_array              FINI_ARRAY        0000000000003dc8   00003dc8
[23] .dynamic                 DYNAMIC            0000000000003dc8   00003dc8
[24] .got                     PROGBITS          0000000000003fb8   00003fb8
[25] .data                    PROGBITS          0000000000004000   00004000
[26] .bss                     PROGBITS          0000000000004000   00004000
[27] .comment                 PROGBITS          0000000000000000   00000000
[28] .symtab                  SYMTAB            0000000000002000   00002000
[29] .strtab                  STRTAB            0000000000003db8   00003db8
[30] .shstrtab                STRTAB            00000000000035dc   000035dc
[31]                         SHSTRTAB          0000000000000000   00000000

Key to Flags:
  V (version), A (alloc), X (executable), M (merge), S (strings), I (info),
  L (link order), O (extra OS processing required), G (group), T (TLS),
  C (compressed), x (contents), o (OS specific), t (tied),
  D (debug), l (large), p (processor specific)
```

- **Lihat program headers**

```
rio_regex@DESKTOP-NVPM5HQ:/tmp/test_dir$ readelf -l program
Elf file type is DYN (Position-Independent Executable file)
Entry point 0x1060
There are 13 program headers, starting at offset 64

Program Headers:
Type           Offset             VirtAddr           PhysAddr
               FileSiz            MemSiz              Flags   Align
-----
PHDR           0x0000000000000040 0x0000000000000040 0x0000000000000040
               R           DxD
INTERP         0x0000000000000318 0x0000000000000318 0x0000000000000318
               R           0x1
               [Requesting program interpreter: /lib64/ld-linux-x86-64.so.2]
LOAD           0x0000000000000000 0x0000000000000000 0x0000000000000000
               R           DxD1000
LOAD           0x0000000000001000 0x0000000000001000 0x0000000000001000
               R           DxD1000
LOAD           0x00000000000001e1 0x00000000000001e1 0x00000000000001e1
               R           DxD1000
LOAD           0x0000000000002000 0x0000000000002000 0x0000000000002000
               R           DxD1000
LOAD           0x000000000000104c 0x000000000000104c 0x000000000000104c
               R           DxD1000
LOAD           0x0000000000002db8 0x0000000000002db8 0x0000000000002db8
               RW          DxD1000
DYNAMIC         0x000000000000258 0x000000000000258 0x000000000000258
               RW          DxD1000
DYNAMIC         0x0000000000003dc8 0x0000000000003dc8 0x0000000000003dc8
               RW          DxD
NOTE           0x0000000000000338 0x0000000000000338 0x0000000000000338
               R           DxD
NOTE           0x0000000000000368 0x0000000000000368 0x0000000000000368
               R           DxD
GNU_PROPERTY    0x0000000000000338 0x0000000000000338 0x0000000000000338
               R           DxD
GNU_EH_FRAME    0x000000000000201c 0x000000000000201c 0x000000000000201c
               R           DxD
GNU_STACK      0x0000000000000000 0x0000000000000000 0x0000000000000000
               RW          DxD10
GNU_RELRO       0x0000000000002db8 0x0000000000002db8 0x0000000000002db8
               R           DxD10
               0x000000000000248 0x000000000000248 0x000000000000248
               R           0x1

Section to Segment mapping:
Segment Sections...
00
01 .interp
02 .interp.note.gnu.property.note.gnu.build-id.note.ABI-tag.gnu.hash.dynsym.dynstr.gnu.version.gnu.version_r.rela.dyn.rela.plt
03 .init.plt.plt.got.plt.sec.text.fini
04 .rodata.eh_frame_hdr.eh_frame
05 .init_array.fini_array.dynamic.got.data.bss
06 .dynamic
07 .note.gnu.property
08 .note.gnu.build-id.note.ABI-tag
09 .note.gnu.property
10 .eh_frame_hdr
11
12 .init_array.fini_array.dynamic.got
```

- Disassemble

```
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ objdump -d program | head -50
program:      file format elf64-x86-64

Disassembly of section .init:

0000000000001000 <.init>:
1000:    f3 0f 1e fa                endbr64
1004:    4b 83 ec 00                sub    $0x0,%rsp
1008:    4b 85 05 c9 2f 00 00      mov    0x2fd9(%rip),%rax        # 3fe8 <__gmon_start__@Base>
100f:    4b 85 c9                    test   %rax,%rax
1012:    74 02                      je     1016 <_init+0x16>
1014:    ff 00                      call   *%rax
1016:    4b 83 c9 08                add    $0x0,%rsp
101a:    c3                          ret

Disassembly of section .plt:

0000000000001020 <.plt>:
1020:    ff 35 9a 2f 00 00        push   0x2f9a(%rip)             # 3fc0 <_GLOBAL_OFFSET_TABLE+0x0>
1026:    ff 25 9c 2f 00 00        jmp     *0x2f9c(%rip)          # 3fc8 <_GLOBAL_OFFSET_TABLE+0x10>
102c:    0f 1f 40 00                nopl    0x0(%rax)
1030:    f3 0f 1e fa                endbr64
1034:    68 00 00 00 00            push    $0x0
1039:    e9 e2 ff ff ff            jmp     1039 <_init+0x20>
103e:    66 90                      xchgb   %ax,%ax

Disassembly of section .plt.got:

0000000000001040 <__cxa_finalize@plt>:
1040:    f3 0f 1e fa                endbr64
1044:    ff 25 aa 2f 00 00        jmp     *0x2faa(%rip)          # 3ff8 <__cxa_finalize@GLIBC_2.2.5>
104a:    66 0f 1f 40 00 00        nopl    0x0(%rax,%rax,1)

Disassembly of section .plt.sec:

0000000000001050 <.plt.sec>:
1050:    f3 0f 1e fa                endbr64
1054:    ff 25 76 2f 00 00        jmp     *0x2f76(%rip)          # 3fd0 <printf@GLIBC_2.2.5>
105a:    66 0f 1f 40 00 00        nopl    0x0(%rax,%rax,1)

Disassembly of section .text:

0000000000001060 <.start>:
1060:    f3 0f 1e fa                endbr64
1064:    31 ed                      xor     %ebp,%ebp
1066:    49 89 d1                    mov     %rdx,%r9
1069:    5e                          pop     %rsi
106a:    4b 89 e2                    mov     %rsp,%rdx
106d:    4b 83 e4 f0                and     $0xffffffffffff0,%rsp
rio_regex@DESKTOP-NVPMSHQ:/tmp/test_dir$ |
```

2.7 OS Design and Implementation

Pada praktik ini, mahasiswa mempelajari bagaimana sistem operasi Linux dirancang dan diimplementasikan melalui **kernel module**. Kernel module merupakan potongan kode yang dapat dimuat (load) dan dilepas (unload) ke dalam kernel tanpa perlu melakukan kompilasi ulang seluruh kernel. Dalam file *hello_module.c*, modul sederhana dibuat untuk menampilkan pesan “Loaded” saat dimasukkan dan “Unloaded” saat dilepas. Proses kompilasi dilakukan dengan *Makefile* menggunakan perintah *make*, lalu modul dimuat ke kernel menggunakan *insmod* dan dihapus dengan *rmmod*. Melalui praktik ini, mahasiswa memahami konsep dasar bagaimana kernel Linux bekerja secara modular, serta cara sistem menangani pesan dan informasi melalui *kernel ring buffer* (*dmesg*). Selain itu, diperkenalkan pula cara memantau dan menganalisis informasi kernel menggunakan perintah seperti *uname -r*, *cat /proc/version*, *sysctl*, *lsmod*, dan *modinfo*. Praktik ini penting untuk memahami bagaimana komponen inti sistem operasi berinteraksi langsung dengan perangkat keras dan proses sistem.

2.7.1 Kernel Module Sederhana

```
rio_regex@DESKTOP-NVPMSHQ:~$ nano hello_module.c
rio_regex@DESKTOP-NVPMSHQ:~$ |
```

- Makefile

```
rio_regex@DESKTOP-NVPMSHQ:~$ sudo nano Makefile
[sudo] password for rio_regex:
rio_regex@DESKTOP-NVPMSHQ:~$ |
```

- **Compile dan load**

```
rio_regex@DESKTOP-NVPM5HQ:~$ make
Makefile:3: *** missing separator. Stop.
rio_regex@DESKTOP-NVPM5HQ:~$ sudo insmod hello_module.ko
insmod: ERROR: could not load module hello_module.ko: No such file or directory
rio_regex@DESKTOP-NVPM5HQ:~$ lsmod | grep hello
rio_regex@DESKTOP-NVPM5HQ:~$ dmesg | tail
[ 2.472290] kvm_amd: Nested Virtualization enabled
[ 2.472291] kvm_amd: Nested Paging enabled
[ 2.472294] kvm_amd: kvm_amd: Hyper-V enlightened NPT TLB flush enabled
[ 2.472296] kvm_amd: kvm_amd: Hyper-V Direct TLB Flush enabled
[ 2.472304] kvm_amd: Virtual VMLOAD VMSAVE supported
[ 4.095353] systemd-journald[41]: /var/log/journal/9735715790fd4c76974e0e8d83320dcc/user-1000.journal: Journal file uses a different sequence number ID, rotating.
[ 6.790723] WSL (239) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 21.585244] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
[ 48.353953] hv_balloon: Max. dynamic memory size: 7806 MB
[ 600.663961] mini_init (224): drop_caches: 1
rio_regex@DESKTOP-NVPM5HQ:~$ sudo rmmod hello_module
rmmod: ERROR: Module hello_module is not currently loaded
rio_regex@DESKTOP-NVPM5HQ:~$ dmesg | tail
[ 2.472290] kvm_amd: Nested Virtualization enabled
[ 2.472291] kvm_amd: Nested Paging enabled
[ 2.472294] kvm_amd: kvm_amd: Hyper-V enlightened NPT TLB flush enabled
[ 2.472296] kvm_amd: kvm_amd: Hyper-V Direct TLB Flush enabled
[ 2.472304] kvm_amd: Virtual VMLOAD VMSAVE supported
[ 4.095353] systemd-journald[41]: /var/log/journal/9735715790fd4c76974e0e8d83320dcc/user-1000.journal: Journal file uses a different sequence number ID, rotating.
[ 6.790723] WSL (239) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 21.585244] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
[ 48.353953] hv_balloon: Max. dynamic memory size: 7806 MB
[ 600.663961] mini_init (224): drop_caches: 1
rio_regex@DESKTOP-NVPM5HQ:~$ |
```

2.7.2 Monitoring Kernel

- **Lihat informasi kernel**

```
rio_regex@DESKTOP-NVPM5HQ:~$ uname -r
6.6.87.2-microsoft-standard-WSL2
rio_regex@DESKTOP-NVPM5HQ:~$ cat /proc/version
Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:30:46 UTC 2025
rio_regex@DESKTOP-NVPM5HQ:~$ |
```

- **Lihat parameter kernel**

```
rio_regex@DESKTOP-NVPM5HQ:~$ sysctl -a | head -20
sysctl: permission denied on key 'kernel.cad_pid'
abi.vsyscall32 = 1
debug.exception-trace = 1
debug.kprobes-optimization = 1
dev.cdrom.autoclose = 1
dev.cdrom.autoeject = 0
dev.cdrom.check_media = 0
dev.cdrom.debug = 0
dev.cdrom.info = CD-ROM information, Id: cdrom.c 3.20 2003/12/17
dev.cdrom.info =
dev.cdrom.info = drive name:
dev.cdrom.info = drive speed:
dev.cdrom.info = drive # of slots:
dev.cdrom.info = Can close tray:
dev.cdrom.info = Can open tray:
dev.cdrom.info = Can lock tray:
dev.cdrom.info = Can change speed:
dev.cdrom.info = Can select disk:
dev.cdrom.info = Can read multisection:
dev.cdrom.info = Can read MCN:
dev.cdrom.info = Reports media changed:
rio_regex@DESKTOP-NVPM5HQ:~$ |
```

- **Lihat kernel ring buffer**

```

rio_regex@DESKTOP-NVPM5HQ:~$ dmesg | tail -30
[ 1.134078] sd 0:0:0:3: Attached scsi generic sg3 type 0
[ 1.134464] sd 0:0:0:3: [sdd] 2147483648 512-byte logical blocks: (1.10 TB/1.00 TiB)
[ 1.135577] sd 0:0:0:3: [sdd] 4096-byte physical blocks
[ 1.136205] sd 0:0:0:3: [sdd] Write Protect is off
[ 1.136855] sd 0:0:0:3: [sdd] Mode Sense: 0f 00 00 00
[ 1.137454] sd 0:0:0:3: [sdd] Write cache: enabled, read cache: enabled, doesn't support DPO or FUA
[ 1.140580] sd 0:0:0:3: [sdd] Attached SCSI disk
[ 1.171749] EXT4-fs (sdd): mounted filesystem 7cc0e06a-568b-4a46-albe-d9b73fc13e50 r/w with ordered data mode. Quota mode: none.
[ 1.792722] pulseaudio[273]: memfd_create() called without MFD_EXEC or MFD_NOEXEC_SEAL set
[ 1.884776] misc dxg: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22
[ 1.888372] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 1.889267] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 1.889662] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 1.890998] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 2.117854] systemd-journald[41]: Collecting audit messages is disabled.
[ 2.180768] systemd-journald[41]: Received client request to flush runtime journal.
[ 2.182820] systemd-journald[41]: File /var/log/journal/9735715790fd4c76974e0e8d83320dcc/system.journal corrupted or uncleanly shut down, renaming and replacing.
[ 2.318317] ACPI: battery: Slot [BAT1] (battery present)
[ 2.326465] ACPI: AC: AC Adapter [AC1] (off-line)
[ 2.472283] kvm_amd: TSC scaling supported
[ 2.472290] kvm_amd: Nested Virtualization enabled
[ 2.472291] kvm_amd: Nested Paging enabled
[ 2.472294] kvm_amd: kvm_amd: Hyper-V enlightened NPT TLB flush enabled
[ 2.472296] kvm_amd: kvm_amd: Hyper-V Direct TLB Flush enabled
[ 2.472304] kvm_amd: Virtual VMLOAD VMSAVE supported
[ 4.095353] systemd-journald[41]: /var/log/journal/9735715790fd4c76974e0e8d83320dcc/user-1000.journal: Journal file uses a different sequence number ID, rotating.
[ 6.790723] WSL (239) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 21.585244] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
[ 48.353953] hv_balloon: Max. dynamic memory size: 7806 MB
[ 600.663961] mini_init (224): drop_caches: 1
rio_regex@DESKTOP-NVPM5HQ:~$

```

- **Lihat kernel modules**

```

rio_regex@DESKTOP-NVPM5HQ:~$ lsmod
Module                  Size  Used by
tls                     98304  0
intel_rapl_msr          16384  0
intel_rapl_common       32768  1 intel_rapl_msr
kvm_amd                 126976  0
kvm                     970752  1 kvm_amd
irqbypass              12288  1 kvm
crc32c_intel            16384  0
ac                      16384  0
battery                20480  0
sch_fq_codel            16384  1
configfs                53248  1
autofs4                 45056  0
br_netfilter            28672  0
bridge                 282624  1 br_netfilter
stp                     12288  1 bridge
llc                     12288  2 bridge, stp
ip_tables               28672  0
tun                     53248  0
rio_regex@DESKTOP-NVPM5HQ:~$

```

- **Informasi modul tertentu**

```

rio_regex@DESKTOP-NVPM5HQ:~$ modinfo ext4
name: ext4
filename: (builtin)
softdep: pre: crc32c
license: GPL
file: fs/ext4/ext4
description: Fourth Extended Filesystem
author: Remy Card, Stephen Tweedie, Andrew Morton, Andreas Dilger, Theodore Ts'o and others
alias: fs-ext4
alias: ext3
alias: fs-ext3
rio_regex@DESKTOP-NVPM5HQ:~$

```

2.8 Operating-System Structure

Pada praktik ini, mahasiswa mempelajari struktur berlapis (*layering structure*) dari sistem operasi Linux serta konsep arsitektur kernel. Linux memiliki lima lapisan utama, yaitu Hardware, Kernel, System Calls, System Programs, dan User Applications. Lapisan hardware dieksplorasi menggunakan perintah seperti lspci, lsusb, dan cat /proc/cpuinfo untuk melihat perangkat yang terpasang. Lapisan kernel diperiksa melalui uname -a, sedangkan lapisan *system calls* diamati dengan strace, yang menampilkan panggilan sistem saat perintah dijalankan. Lapisan *system programs* seperti ps aux dan top menunjukkan proses aktif, sementara *user applications* seperti bash dan gcc mewakili aplikasi pengguna.

Pada praktik Microkernel vs Monolithic, mahasiswa membandingkan ukuran kernel Linux dan mempelajari perbedaan antara fitur *built-in* dan *modular* menggunakan `ls -lh`, `grep`, serta `lsmod`. Praktik ini menunjukkan bahwa Linux menggunakan pendekatan monolithic kernel, di mana sebagian besar fungsi dijalankan di ruang kernel.

Selanjutnya, dalam praktik Virtual File System (VFS), program `procfs_read.c` dibuat untuk membaca informasi sistem dari direktori `/proc`, seperti data CPU, memori, dan proses yang sedang berjalan. Hal ini menunjukkan bagaimana Linux menggunakan *virtual file system* untuk menyediakan antarmuka fleksibel antara kernel dan aplikasi pengguna tanpa langsung mengakses perangkat keras. Melalui praktik ini, mahasiswa memahami bagaimana struktur sistem operasi diorganisasikan dan berinteraksi secara hierarkis dari level perangkat keras hingga aplikasi pengguna.

2.8.1 Eksplorasi Struktur Layering

- Layer 1: Hardware

```
rio_regex@DESKTOP-NVPM5HQ:~$ lspci
Command 'lspci' not found, but can be installed with:
sudo apt install pciutils
rio_regex@DESKTOP-NVPM5HQ:~$ lsusb
Command 'lsusb' not found, but can be installed with:
sudo apt install usbutils
rio_regex@DESKTOP-NVPM5HQ:~$ cat /proc/cpuinfo
processor       : 0
vendor_id      : AuthenticAMD
cpu family     : 25
model          : 68
model name     : AMD Ryzen 5 6600H with Radeon Graphics
stepping       : 1
microcode      : 0xffffffff
cpu MHz        : 3293.811
cache size     : 512 KB
physical id    : 0
siblings       : 12
core id        : 0
cpu cores      : 6
apicid         : 0
initial apicid : 0
fpu            : yes
fpu_exception  : yes
cpuid level    : 13
wp             : yes
flags          : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtscp lm constab
nt_tsc rep_good nopl tsc_reliable nonstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor
lahf_lm cmp_legacy svm cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw topoext perfctr_core ssbd ibrs ibpb stibp vmcall fsgsbase bmi1 avx2 smep bmi2 erms invpcid r
dseed adx aomp clflushopt clmb sha_ni xsaveopt xsavec xgetbv1 xsaves clzero xsaveerptr arat npt nrip_save tsc_scale vmcb_clean flushbyasid decodeassists pausefilter pfthr
eshold v_vmsave_vmload umip vaes vpclmulqdq rdpid fsrm
bugs           : sysret_ss_attrs null_seg spectre_v1 spectre_v2 spec_store_bypass srso ibpb_no_ret
bogomips       : 6587.62
tlb size       : 2560 4K pages
clflush size   : 64
cache_alignmen : 64
address sizes   : 48 bits physical, 48 bits virtual
power management:

processor       : 1
vendor_id      : AuthenticAMD
cpu family     : 25
model          : 68
model name     : AMD Ryzen 5 6600H with Radeon Graphics
stepping       : 1
microcode      : 0xffffffff
cpu MHz        : 3293.811
cache size     : 512 KB
physical id    : 0
siblings       : 12
```

- Layer 2: Kernel

```
rio_regex@DESKTOP-NVPM5HQ:~$ uname -a
Linux DESKTOP-NVPM5HQ 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
rio_regex@DESKTOP-NVPM5HQ:~$ cat /proc/sys/kernel/ostype
Linux
rio_regex@DESKTOP-NVPM5HQ:~$ |
```

- Layer 3: System Calls

```
rio_regex@DESKTOP-NVPM5HQ:~$ strace ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
rio_regex@DESKTOP-NVPM5HQ:~$ |
```

- **Layer 4: System Programs**

```

rio_regex@DESKTOP-NVPM5HQ:~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.1 21672 12244 ?        Ss   16:47  0:00 /sbin/init
root         2  0.0  0.0   3072  1792 ?        Ss   16:47  0:00 /init
root         7  0.0  0.0   3072  1792 ?        Ss   16:47  0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-trunc
root        41  0.0  0.2 66820 16520 ?        Ss   16:47  0:00 /usr/lib/systemd/systemd-journald
root        91  0.0  0.0 25808  6400 ?        Ss   16:47  0:00 /usr/lib/systemd/systemd-udev
systemd+  157  0.0  0.1 21456 12672 ?        Ss   16:47  0:00 /usr/lib/systemd/systemd-resolved
systemd+  161  0.0  0.0 91824  7680 ?        Ss   16:47  0:00 /usr/lib/systemd/systemd-timesyncd
root       169  0.0  0.0   4236  2560 ?        Ss   16:47  0:00 /usr/sbin/cron -- -p
message+  170  0.0  0.0   9632  4736 ?        Ss   16:47  0:00 dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activ
root       177  0.0  0.1 17960  8448 ?        Ss   16:47  0:00 /usr/lib/systemd/systemd-logind
root       179  0.0  0.1 1755584 12160 ?       Ss   16:47  0:00 /usr/libexec/ssl-pro-service -vv
root       182  0.0  0.0   2160  1920 hvc0     S+   16:47  0:00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
syslog    193  0.0  0.0 222508  5376 ?        Ss   16:47  0:00 /usr/sbin/rsyslogd -n -iNONE
root      201  0.0  0.0   3116  1792 tty1     S+   16:47  0:00 /sbin/agetty -o -p -- \u --noclear - linux
root      210  0.0  0.2 107032 22400 ?        Ss   16:47  0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown -
root      300  0.0  0.0   6664  4352 pts/1    Ss   16:47  0:00 /bin/login -f
rio_regex 345  0.0  0.1 20308 10880 ?        Ss   16:47  0:00 /usr/lib/systemd/systemd --user
rio_regex 347  0.0  0.0 21156  3520 ?        S   16:47  0:00 (sd-pam)
rio_regex 366  0.0  0.0   6072  5248 pts/1    S+   16:47  0:00 -bash
root      474  0.0  0.0   3076  1032 ?        Ss   16:47  0:00 /init
root      475  0.0  0.0   3092  1164 ?        Ss   16:47  0:00 /init
rio_regex 481  0.0  0.0   6072  5248 pts/0    Ss   16:47  0:00 -bash
rio_regex 618  0.0  0.0   8332  4224 pts/0    R+   17:39  0:00 ps aux

rio_regex@DESKTOP-NVPM5HQ:~$ top
top - 17:40:02 up 52 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 23 total, 1 running, 22 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.1 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7861.2 total,  7003.3 free,  537.9 used,  169.0 buff/cache
MiB Swap: 2048.0 total,  2048.0 free,  0.0 used,  7023.3 avail Mem

   PID USER      PR  NI   VIRT  RES  SHR  S  %CPU  %MEM     TIME+ COMMAND
    618 rio_regex 20   0   8336   3760 3456 R   15   0.1   0:00.00 top
      1 root      20   0  21672 12244 9428 S   0.0  0.2   0:00.61 systemd
      2 root      20   0   3072  1792  1792 S   0.0  0.0   0:00.01 init-systemd(Ub
      7 root      20   0   3072  1792  1792 S   0.0  0.0   0:00.00 init
     41 root     19  -1 66820 16520 15752 S   0.0  0.2   0:00.20 systemd-journal
     91 root      20   0 25808  6400  4992 S   0.0  0.1   0:00.27 systemd-udev
    157 systemd+  20   0 21456 12672 10496 S   0.0  0.2   0:00.11 systemd-resolve
    161 systemd+  20   0 91824  7680  6784 S   0.0  0.1   0:00.14 systemd-timesyn
    169 root      20   0   4236  2560  2432 S   0.0  0.0   0:00.01 cron
    170 message+  20   0   9632  4736  4352 S   0.0  0.1   0:00.07 dbus-daemon
    177 root      20   0 17960  8448  7552 S   0.0  0.1   0:00.09 systemd-logind
    179 root      20   0 1755584 12160 10240 S   0.0  0.2   0:00.17 ssl-pro-service
    182 root      20   0   2160  1920  1792 S   0.0  0.0   0:00.01 agetty
    193 syslog    20   0 222508  5376  4352 S   0.0  0.1   0:00.08 rsyslogd
    201 root      20   0   3116  1792  1664 S   0.0  0.0   0:00.01 agetty
    210 root      20   0 107032 22400 13184 S   0.0  0.3   0:00.09 unattended-upgr
    300 root      20   0   6664  4352  4352 S   0.0  0.1   0:00.01 login
    345 rio_regex 20   0 20308 10880  8960 S   0.0  0.1   0:00.09 systemd
    347 rio_regex 20   0 21156  3520  1792 S   0.0  0.0   0:00.00 (sd-pam)
    366 rio_regex 20   0   6072  5248  3584 S   0.0  0.1   0:00.02 bash
    474 root      20   0   3076  1032   896 S   0.0  0.0   0:00.00 sessionLeader
    475 root      20   0   3092  1164  1024 S   0.0  0.0   0:00.25 Relay(481)

```

- **Layer 5: User Applications**

```

rio_regex@DESKTOP-NVPM5HQ:~$ User Applications
w
User: command not found
18:12:35 up 1:04, 1 user, load average: 0.00, 0.00, 0.00
USER      TTY      FROM          LOGIN@   IDLE   JCPU   PCPU   WHAT
rio_regex pts/1    -            16:47    1:24m  0.03s  0.02s -bash
rio_regex@DESKTOP-NVPM5HQ:~$ which gcc
/usr/bin/gcc
rio_regex@DESKTOP-NVPM5HQ:~$

```

2.8.2 Microkernel vs Monolithic

- **Lihat ukuran kernel (Monolithic)**

```

rio_regex@DESKTOP-NVPM5HQ:~$ ls -lh /boot/vmlinuz-$(uname -r)
ls: cannot access '/boot/vmlinuz-6.6.87.2-microsoft-standard-WSL2': No such file or directory
rio_regex@DESKTOP-NVPM5HQ:~$

```

- **Lihat built-in vs module**

```

rio_regex@DESKTOP-NVPM5HQ:~$ cat /boot/config-$(uname -r) | grep "=y" | wc -l
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
rio_regex@DESKTOP-NVPM5HQ:~$ cat /boot/config-$(uname -r) | grep "=m" | wc -l
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
rio_regex@DESKTOP-NVPM5HQ:~$

```

- **Lihat loaded modules**

```

rio_regex@DESKTOP-NVPM5HQ:~$ lsmod | wc -l
18
rio_regex@DESKTOP-NVPM5HQ:~$

```

2.8.3 Program - Virtual File System

```
rio_regex@DESKTOP-NVPMSHQ:~$ nano procfs_read.c
rio_regex@DESKTOP-NVPMSHQ:~$ gcc -o procfs_read procfs_read.c
procfs_read.c: In function 'main':
procfs_read.c:6:11: error: expected ';' before 'printf'
  6 | 2025-10-06
    | ^
    | ;
    | // Baca informasi CPU dari /proc
    | 8 | printf("=== CPU Information ===\n");
procfs_read.c:30:59: warning: implicit declaration of function 'getpid' [-Wimplicit-function-declaration]
   30 | snprintf(proc_path, sizeof(proc_path), "/proc/%d/status", getpid());
      |                                           ^~~~~~
rio_regex@DESKTOP-NVPMSHQ:~$ ./procfs_read
-bash: ./procfs_read: No such file or directory
rio_regex@DESKTOP-NVPMSHQ:~$ |
```

2.9 Building and Booting an Operating System

Pada praktik ini, mahasiswa mempelajari bagaimana sistem operasi Linux dibangun dan dijalankan melalui tahapan proses booting serta pengaturan kernel parameters. Proses booting merupakan langkah awal ketika komputer dinyalakan, dimulai dari BIOS/UEFI, dilanjutkan ke bootloader (GRUB), lalu memuat kernel dan sistem utama hingga siap digunakan.

Melalui perintah `dmesg | less`, mahasiswa dapat melihat pesan kernel saat sistem melakukan inisialisasi perangkat keras. Kemudian, dengan `systemd-analyze`, `systemd-analyze blame`, dan `systemd-analyze critical-chain`, mahasiswa dapat menganalisis waktu yang dibutuhkan setiap layanan untuk memulai, serta mengidentifikasi proses yang paling kritis dalam booting. File konfigurasi GRUB (`/boot/grub/grub.cfg`) memperlihatkan urutan dan parameter yang digunakan saat sistem dijalankan, sedangkan `lsinitramfs` menampilkan isi dari *initial RAM filesystem* (`initramfs`) yang berisi driver dan skrip penting untuk menyiapkan sistem sebelum root filesystem dimuat.

Pada praktik Kernel Parameters, mahasiswa mempelajari cara membaca dan mengubah parameter kernel yang digunakan saat boot. Perintah `cat /proc/cmdline` menunjukkan parameter aktif, sedangkan untuk mengubahnya dapat dilakukan secara sementara (dengan menekan `e` pada menu GRUB saat boot) atau permanen dengan mengedit file `/etc/default/grub` dan menjalankan `sudo update-grub`.

Praktik ini membantu mahasiswa memahami urutan kerja sistem operasi dari awal komputer menyala hingga siap digunakan, serta bagaimana pengaturan kernel dapat memengaruhi performa dan perilaku sistem secara keseluruhan.

2.9.1 Analisis Boot Process

- **Lihat boot messages**

```
rio_regex@DESKTOP-NVPMSHQ:~$ dmesg | less
[1]+  Stopped                  dmesg | less
rio_regex@DESKTOP-NVPMSHQ:~$ |
```

- **Lihat systemd boot analysis**

```

fio_regex@DESKTOP-NVPM5HQ:~$ systemctl analyze
Startup finished in 1.306s (userspace)
graphical.target reached after 1.324s in userspace.
fio_regex@DESKTOP-NVPM5HQ:~$ systemctl analyze blame
672ms landscape-client.service
298ms snapd.seeded.service
299ms dev-mtd.device
199ms snapd.service
152ms user@1000.service
183ms systemd-udev-trigger.service
143ms wsl-pro.service
136ms snapd.socket
123ms systemd-resolved.service
75ms rsyslog.service
70ms systemd-timesyncd.service
55ms systemd-journal-flush.service
52ms keyboard-setup.service
50ms systemd-journald.service
49ms systemd-logind.service
41ms systemd-udevd.service
35ms user-runtime-dir@1000.service
35ms s2scrub_reap.service
22ms systemd-tmpfiles-setup.service
30ms dbus.service
24ms dev-hugepages.mount
24ms dev-mqueue.mount
23ms sys-kernel-debug.mount
22ms sys-kernel-tracing.mount
21ms systemd-sysctl.service
19ms systemd-tmpfiles-setup-dev-early.service
18ms systemd-update-utmp.service
17ms systemd-bifrost.service
17ms kmod-static-nodes.service
16ms modprobe@config.service
16ms modprobe@kmod.service
16ms modprobe@dm.service
15ms modprobe@efl.service
15ms modprobe@fuse.service
15ms systemd-tmpfiles-setup-dev.service
14ms modprobe@loop.service
lines 1-33
[1] * Stopped
fio_regex@DESKTOP-NVPM5HQ:~$ [200-systemd-analyze critical-chain=
[200-systemd-analyze: command not found
fio_regex@DESKTOP-NVPM5HQ:~$ systemctl analyze critical-chain
The time when unit became active or started is printed after the "0" character.
The time the unit took to start is printed after the "+" character.
graphical.target @1.324s
├─multiuser.target @1.322s
│   └─snapd.seeded.service @57ms +298ms
│       └─basic.target @64ms
│           └─sockets.target @64ms
│               └─snapd.socket @50ms +136ms
│                   └─systemd.target @60ms
│                       └─systemd-resolved.service @370ms +123ms
│                           └─systemd-tmpfiles-setup.service @23ms +32ms
│                               └─systemd-journal-flush.service @272ms +55ms
│                                   └─systemd-journald.service @220ms +50ms
│                                       └─systemd-journald.socket @212ms
│                                           └─system.slice @170ms
│                                               └─system.slice @175ms
fio_regex@DESKTOP-NVPM5HQ:~$

```

- **Lihat GRUB configuration**

```

fio_regex@DESKTOP-NVPM5HQ:~$ cat /boot/grub/grub.cfg | head -50
cat: /boot/grub/grub.cfg: No such file or directory
fio_regex@DESKTOP-NVPM5HQ:~$

```

- **Lihat initramfs**

```

fio_regex@DESKTOP-NVPM5HQ:~$ lsinitramfs /boot/initrd.img-$(uname -r) | head -20
Command 'lsinitramfs' not found, but can be installed with:
sudo apt install initramfs-tools-core
fio_regex@DESKTOP-NVPM5HQ:~$

```

2.10 Operating-System Debugging

Praktik Operating-System Debugging pada Linux bertujuan untuk memahami cara menemukan dan menganalisis kesalahan dalam sistem maupun program. Dengan strace, pengguna dapat melacak system calls yang dijalankan oleh perintah seperti ls, serta melihat berapa lama setiap system call berlangsung. Sementara itu, ltrace digunakan untuk memantau panggilan library functions agar programmer mengetahui fungsi-fungsi pustaka apa saja yang dijalankan. GDB (GNU Debugger) berfungsi mendeteksi kesalahan logika atau bug dalam program, seperti pada contoh buggy.c yang memiliki kesalahan pada batas array. dmesg digunakan untuk membaca log kernel, memantau error, atau peringatan sistem. Selain itu, Valgrind membantu mendeteksi kebocoran memori, sedangkan perf digunakan untuk menganalisis performa sistem dan memonitor aktivitas CPU. Semua alat ini memungkinkan pengguna memahami perilaku sistem operasi secara mendalam dan memperbaiki masalah dengan efisien.

2.10.1 Using strace

- **Trace system calls**

[illegible]

- **Trace dengan detail**

[illegible]

- **Hitung waktu system calls**

```

rio_regex@DESKTOP-NVPM5HQ:~$ strace -c ls
Makefile      file_monitor.c  ipc_benchmark  os_services    process_basic   process_manager.c  projects  shm_reader.c  shmfile
fifo_reader.c  fork_demo.c     ipc_benchmark.c os_services.c   process_basic.c  process_wait.c     reader    shm_writer    test_file.txt
fifo_writer.c  hello_module.c  multiple_fork.c pipe_demo.c     process_manager  procfs_read.c     shm_reader shm_writer.c   writer
% time        seconds    usecs/call      multiple calls  errors syscalls
-----
21.09 0.0000004      25      35      13 openat
19.25 0.0000025      27      38
13.72 0.000588      588      1      execve
10.88 0.000432      43      10      read
9.96 0.000427      17      24      close
9.19 0.000394      17      23      fstat
3.52 0.000151      50      3      write
2.99 0.000128      25      5      mprotect
1.45 0.000062      31      2      getdents64
1.35 0.000058      29      2      1 statfs
1.10 0.000047      15      3      bmk
1.00 0.000043      21      2      2 access
0.91 0.000039      19      2      ioctl
0.79 0.000034      17      2      pread64
0.72 0.000031      31      1      munmap
0.70 0.000030      30      1      futex
0.37 0.000016      16      1      arch_prctl
0.37 0.000016      16      1      set_robust_list
0.37 0.000016      16      1      prlimit64
0.35 0.000015      15      1      set_tid_address
0.35 0.000015      15      1      prlimit64
0.35 0.000015      15      1      rseq
100.00 0.004286      28      152      16 total
rio_regex@DESKTOP-NVPM5HQ:~$

```

- **Trace dan simpan ke file**

```

rio_exegeth@DESKTOP-NVPM5HQ:~$ strace -o trace.txt ls
Makefile      file_monitor.c  ipc_benchmark  os_services.c  process_basic.c  process_manager.c  projects      shm_reader.c  shmfile      writer
fifo_reader.c  fork_demo.c    ipc_benchmark.c  os_services.c  process_basic.c  process_wait.c    reader        shm_writer    test_file.txt
fifo_writer.c  hello_module.c  multiple_fork.c  pipe_demo.c    process_manager.c  procfs_read.c     shm_writer.c  trace.txt

rio_exegeth@DESKTOP-NVPM5HQ:~$ cat trace.txt
execve("/usr/bin/ls", ["ls"], 0x7fff0581820 /* 26 vars */) = 0
brk(NULL) = 0x5daf2f055000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7975339d6000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20919, ...}) = 0
mmap(NULL, 20919, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7975339d1000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=174472, ...}) = 0
mmap(NULL, 181960, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7975339a4000
mmap(0x7975339aa000, 118784, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x6000) = 0x7975339aa000
mmap(0x7975339c7000, 24576, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x23000) = 0x7975339c7000
mmap(0x7975339cd000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x29000) = 0x7975339cd000
mmap(0x7975339cf000, 5832, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7975339cf000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0", 832) = 832
pread64(3, "\6\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0", 784, 64) = 784
fstat(3, {st_mode=S_IFREG|0755, st_size=2125328, ...}) = 0
pread64(3, "\6\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0", 784, 64) = 784
mmap(NULL, 2170256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x797533600000
mmap(0x797533628000, 1605632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x797533628000
mmap(0x7975337b0000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1b0000) = 0x7975337b0000
mmap(0x7975337ff000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fe000) = 0x7975337ff000
mmap(0x797533805000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x797533805000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=623344, ...}) = 0
mmap(NULL, 627472, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7975339a0000
mmap(0x7975339ac000, 405660, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) = 0x7975339ac000
mmap(0x79753399a000, 163800, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x70000) = 0x79753399a000
mmap(0x7975339a2000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x97000) = 0x7975339a2000
close(3) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x797533907000
arch_prctl(ARCH_SET_FS, 0x797533907000) = 0
set_tid_address(0x797533907ad0) = 435
set_robust_list(0x797533907ae0, 24) = 0

```

2.10.2 Using ltrace

- **Trace library calls**

```
rio_regex@DESKTOP-NVPM5HQ:~$ ltrace ls
Command 'ltrace' not found, but can be installed with:
sudo apt install ltrace
rio_regex@DESKTOP-NVPM5HQ:~$ |
```

- **Dengan detail**

```
rio_regex@DESKTOP-NVPM5HQ:~$ ltrace -c ls
Command 'ltrace' not found, but can be installed with:
sudo apt install ltrace
rio_regex@DESKTOP-NVPM5HQ:~$ |
```

2.10.3 Using GDB

```

rio_regex@DESKTOP-NVPM5HQ:~$ nano buggy.c
rio_regex@DESKTOP-NVPM5HQ:~$

```

- Compile dengan debug symbols

```
rio_regex@DESKTOP-NVPM5HQ:~$ gcc -g -o buggy buggy.c
rio_regex@DESKTOP-NVPM5HQ:~$
```

- Jalankan dengan GDB

```
rio_regex@DESKTOP-NVPM5HQ:~$ gdb ./buggy
GNU gdb (Ubuntu 15.0.50.20240403-0ubuntu1) 15.0.50.20240403-git
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
<http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./buggy...
(gdb)
```

- Debug dengan GDB

```
This GDB supports auto-downloading debuginfo from the following URLs:
  <https://debuginfod.ubuntu.com>
Enable debuginfod for this session? (y or [n]) y
Debuginfod has been enabled.
To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.
Downloading separate debug info for system-supplied DSO at 0x7ffff7fc3000
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at buggy.c:14
14      buggy_function(5);
(gdb) next
array[0] = 0
array[1] = 2
array[2] = 4
array[3] = 6
array[4] = 8
15      return 0;
(gdb) step
16      }
(gdb) print n
No symbol "n" in current context.
(gdb) continue
Continuing.
[Inferior 1 (process 427) exited normally]
(gdb) backtrace
No stack.
(gdb) Quit
(gdb) quit
rio_regex@DESKTOP-NVPM5HQ:~$
```

2.10.4 Kernel Debugging dengan dmesg

- Lihat kernel log

```
root@kali:~# dmesg
Linux version 6.6.0-2-microsoft-standard-hls1 (root@39a258ad54w) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:38:46 UTC 2025
Command line: initrd=initrd.img WSL_ROOT_INIT= panic=-1 nr_cpus=12 hv_utils.timesync_implicit=1 console=hvc0 debug ptty_legacy_count=8 WSL_ENABLE_CRASH_DUMP=1
KERNEL supported cpus:
Intel GenuineIntel
AMD AuthenticAMD
BIOS-provided physical RAM map:
BIOS-e820: [mem 0x0000000000000000-0x00000000000009ffff] usable
BIOS-e820: [mem 0x0000000000000900-0x0000000000000e9fff] reserved
BIOS-e820: [mem 0x0000000000100000-0x00000000000000ffff] ACPI data
BIOS-e820: [mem 0x0000000000200000-0x00000000000000ffff] usable
BIOS-e820: [mem 0x0000000010000000-0x0000000010000000] usable
NX (Execute Disable) protection: active
APIC: Static calls initialized
DMI not present or invalid.
Hypervisor detected: Microsoft Hyper-V
Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0x900c2c, misc 0xe0bed7b6
Hyper-V: Nested features: 0x4a0000
Hyper-V: LAPIC Timer Frequency: 0xc3500
Hyper-V: Using hypercall for remote TSB flush
clocksource: hyperv_clocksource.tsc_page: mask: 0xffffffffffffff max_cycles: 0x246e6a1710, max_idle_ns: 440795202120 ns
clocksource: hyperv_clocksource.nr: mask: 0xffffffffffffff max_cycles: 0x246e6a1710, max_idle_ns: 440795202120 ns
tsc: Detected 3293.811 Mhz processor
e820: update [mem 0x00000000-0x000009ff] usable ==> reserved
e820: remove [mem 0x00000900-0x000009ff] usable
Last_pfn = 0x1ef00 max_arch_pfn = 0x00000000
MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
Last_pfn = 0xf000 max_arch_pfn = 0x00000000
Using GB pages for direct mapping
RAMDISK: [mem 0x04000000-0x04000000]
ACPI: Early table checksum verification disabled
ACPI: RSDP 0x0000000000000000 000024 (v02 VIRTUAL)
ACPI: XSDT 0x0000000000100000 000044 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
ACPI: FACP 0x0000000000100000 000114 (v06 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
ACPI: DSDT 0x00000000001011B8 01E11C (v02 MSFTVM DSDT01 00000001 INTL 20230628)
ACPI: FACS 0x00000000001011A4 000040
ACPI: OEM0 0x0000000000101154 000064 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
ACPI: SRAT 0x000000000011F2D4 000370 (v02 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
ACPI: APIC 0x000000000011F644 0000A8 (v04 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
ACPI: Reserving FACP table memory at [mem 0x101000-0x101133]
ACPI: Reserving DSDT table memory at [mem 0x1011B8-0x11F2D3]
ACPI: Reserving FACS table memory at [mem 0x101114-0x101153]
ACPI: Reserving OEM0 table memory at [mem 0x101154-0x1011B7]
ACPI: Reserving SRAT table memory at [mem 0x11F2D4-0x11F643]
ACPI: Reserving APIC table memory at [mem 0x11F644-0x11F66B]
SRAT: PXM 0 -> APIC 0x00 -> Node 0
SRAT: PXM 0 -> APIC 0x01 -> Node 0
SRAT: PXM 0 -> APIC 0x02 -> Node 0
SRAT: PXM 0 -> APIC 0x03 -> Node 0
SRAT: PXM 0 -> APIC 0x04 -> Node 0
SRAT: PXM 0 -> APIC 0x05 -> Node 0
SRAT: PXM 0 -> APIC 0x06 -> Node 0
SRAT: PXM 0 -> APIC 0x07 -> Node 0
SRAT: PXM 0 -> APIC 0x08 -> Node 0
```

- Follow kernel log

```
root@kali:~# dmesg -w
Linux version 6.6.0-2-microsoft-standard-hls1 (root@39a258ad54w) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:38:46 UTC 2025
Command line: initrd=initrd.img WSL_ROOT_INIT= panic=-1 nr_cpus=12 hv_utils.timesync_implicit=1 console=hvc0 debug ptty_legacy_count=8 WSL_ENABLE_CRASH_DUMP=1
KERNEL supported cpus:
Intel GenuineIntel
AMD AuthenticAMD
BIOS-provided physical RAM map:
BIOS-e820: [mem 0x0000000000000000-0x00000000000009ffff] usable
BIOS-e820: [mem 0x0000000000000900-0x0000000000000e9fff] reserved
BIOS-e820: [mem 0x0000000000100000-0x00000000000000ffff] ACPI data
BIOS-e820: [mem 0x0000000000200000-0x00000000000000ffff] usable
BIOS-e820: [mem 0x0000000010000000-0x0000000010000000] usable
NX (Execute Disable) protection: active
APIC: Static calls initialized
DMI not present or invalid.
Hypervisor detected: Microsoft Hyper-V
Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0x900c2c, misc 0xe0bed7b6
Hyper-V: Nested features: 0x4a0000
Hyper-V: LAPIC Timer Frequency: 0xc3500
Hyper-V: Using hypercall for remote TSB flush
clocksource: hyperv_clocksource.tsc_page: mask: 0xffffffffffffff max_cycles: 0x246e6a1710, max_idle_ns: 440795202120 ns
clocksource: hyperv_clocksource.nr: mask: 0xffffffffffffff max_cycles: 0x246e6a1710, max_idle_ns: 440795202120 ns
tsc: Detected 3293.811 Mhz processor
e820: update [mem 0x00000000-0x000009ff] usable ==> reserved
e820: remove [mem 0x00000900-0x000009ff] usable
Last_pfn = 0x1ef00 max_arch_pfn = 0x00000000
MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
Last_pfn = 0xf000 max_arch_pfn = 0x00000000
Using GB pages for direct mapping
RAMDISK: [mem 0x04000000-0x04000000]
ACPI: Early table checksum verification disabled
ACPI: RSDP 0x0000000000000000 000024 (v02 VIRTUAL)
ACPI: XSDT 0x0000000000100000 000044 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
ACPI: FACP 0x0000000000100000 000114 (v06 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
ACPI: DSDT 0x00000000001011B8 01E11C (v02 MSFTVM DSDT01 00000001 INTL 20230628)
ACPI: FACS 0x00000000001011A4 000040
ACPI: OEM0 0x0000000000101154 000064 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
ACPI: SRAT 0x000000000011F2D4 000370 (v02 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
ACPI: APIC 0x000000000011F644 0000A8 (v04 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
ACPI: Reserving FACP table memory at [mem 0x101000-0x101133]
ACPI: Reserving DSDT table memory at [mem 0x1011B8-0x11F2D3]
ACPI: Reserving FACS table memory at [mem 0x101114-0x101153]
ACPI: Reserving OEM0 table memory at [mem 0x101154-0x1011B7]
ACPI: Reserving SRAT table memory at [mem 0x11F2D4-0x11F643]
ACPI: Reserving APIC table memory at [mem 0x11F644-0x11F66B]
SRAT: PXM 0 -> APIC 0x00 -> Node 0
SRAT: PXM 0 -> APIC 0x01 -> Node 0
SRAT: PXM 0 -> APIC 0x02 -> Node 0
SRAT: PXM 0 -> APIC 0x03 -> Node 0
SRAT: PXM 0 -> APIC 0x04 -> Node 0
SRAT: PXM 0 -> APIC 0x05 -> Node 0
SRAT: PXM 0 -> APIC 0x06 -> Node 0
SRAT: PXM 0 -> APIC 0x07 -> Node 0
SRAT: PXM 0 -> APIC 0x08 -> Node 0
```

- Filter by level

```
rio_regex@DESKTOP-NVPM5HQ:~$ dmesg -l err
[ 0.103564] PCI: Fatal: No config space access function found
[ 1.672701] misc dkg: dkgio_is.feature.enabled: ioctl failed: -22
[ 1.677041] misc dkg: dkgio_query_adapter_info: ioctl failed: -22
[ 1.678580] misc dkg: dkgio_query_adapter_info: ioctl failed: -22
[ 1.688104] misc dkg: dkgio_query_adapter_info: ioctl failed: -22
[ 1.682203] misc dkg: dkgio_query_adapter_info: ioctl failed: -2
[ 6.418168] WOL (235) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 499.379791] misc dkg: dkgio_is.feature.enabled: ioctl failed: -22
[ 499.384646] misc dkg: dkgio_query_adapter_info: ioctl failed: -22
[ 499.385277] misc dkg: dkgio_query_adapter_info: ioctl failed: -22
[ 499.389702] misc dkg: dkgio_query_adapter_info: ioctl failed: -22
[ 499.384815] misc dkg: dkgio_query_adapter_info: ioctl failed: -2
[ 3286.405910] misc dkg: dkgio_is.feature.enabled: ioctl failed: -22
[ 3286.406886] misc dkg: dkgio_query_adapter_info: ioctl failed: -22
[ 3286.406790] misc dkg: dkgio_query_adapter_info: ioctl failed: -22
[ 3286.406632] misc dkg: dkgio_query_adapter_info: ioctl failed: -22
[ 3286.407613] misc dkg: dkgio_query_adapter_info: ioctl failed: -2
rio_regex@DESKTOP-NVPM5HQ:~$ dmesg -l warn
[ 0.678429] Speculative Return Stack Overflow: IDPB-extending microcode not applied!
[ 0.678429] Speculative Return Stack Overflow: WARNING: See https://kernel.org/doc/htm/latest/admin-guide/hw-vuln/srso.html for mitigation options.
[ 0.134696] PCI: System does not support PCI
[ 0.247946] device-mapper: core: CONFIG_IMA_DISABLE_HTABLE is disabled. Duplicate IMA measurements will not be recorded in the IMA log.
[ 1.576092] pulseaudio[270]: memfd_create() called without MFD_EXEC or MFD_NOEXEC_SEAL set
[ 1.919902] systemd-journal[41]: File /var/log/journal/9735715790f4dc76974e8e8d8332bdc/system.journal corrupted or uncleanly shut down, renaming and replacing.
[ 3.767832] systemd-journal[41]: File /var/log/journal/9735715790f4dc76974e8e8d8332bdc/user-1000.journal corrupted or uncleanly shut down, renaming and replacing.
[ 409.440151] Exception:
[ 409.440164] Operation canceled @p9io.cpp:258 (AcceptAsync)
[ 499.855168] systemd-journal[41]: File /var/log/journal/9735715790f4dc76974e8e8d8332bdc/system.journal corrupted or uncleanly shut down, renaming and replacing.
[ 502.142110] systemd-journal[41]: File /var/log/journal/9735715790f4dc76974e8e8d8332bdc/user-1000.journal corrupted or uncleanly shut down, renaming and replacing.
[ 507.536350] ICP: eth0: Driver has suspect GRO implementation, ICP performance may be compromised.
[ 3283.913685] Exception:
[ 3283.913750] Operation canceled @p9io.cpp:258 (AcceptAsync)
[ 3286.781630] Failed to connect to bus: No such file or directory
[ 3286.950109] systemd-journal[41]: File /var/log/journal/9735715790f4dc76974e8e8d8332bdc/system.journal corrupted or uncleanly shut down, renaming and replacing.
rio_regex@DESKTOP-NVPM5HQ:~$
```

2.10.5 Memory Leak Detection dengan Valgrind

- Install valgrind

```
rio_regex@DESKTOP-NVPM5HQ:~$ sudo apt install valgrind
[sudo] password for rio_regex:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
valgrind is already the newest version (1:3.22.0-0ubuntu3).
0 upgraded, 0 newly installed, 0 to remove and 28 not upgraded.
rio_regex@DESKTOP-NVPM5HQ:~$
```

- Buat program dengan memory leak

```
rio_regex@DESKTOP-NVPM5HQ:~$ cat > memleak.c << 'EOF'
> #include <stdlib.h>
int main() {
    int *ptr = (int*)malloc(100 * sizeof(int));
    // Forgot to free(ptr)
return 0;
}
EOF
gcc -g -o memleak memleak.c #include <stdlib.h>
int main() {
    int *ptr = (int*)malloc(100 * sizeof(int));
    // Forgot to free(ptr)
return 0;
}
EOF^C
rio_regex@DESKTOP-NVPM5HQ:~$ cat > memleak.c << 'EOF'
> #include <stdlib.h>
int main() {
    int *ptr = (int*)malloc(100 * sizeof(int));
    // Forgot to free(ptr)
return 0;
}
EOF
rio_regex@DESKTOP-NVPM5HQ:~$ gcc -g -o memleak memleak.c
```

- Jalankan dengan valgrind

```
rio_regex@DESKTOP-NVPM5HQ:~$ valgrind --leak-check=full ./memleak
==630== Memcheck, a memory error detector
==630== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==630== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==630== Command: ./memleak
==630==
==630== HEAP SUMMARY:
==630==   in use at exit: 400 bytes in 1 blocks
==630==   total heap usage: 1 allocs, 0 frees, 400 bytes allocated
==630==
==630== 400 bytes in 1 blocks are definitely lost in loss record 1 of 1
==630==   at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==630==   by 0x10915E: main (memleak.c:3)
==630==
==630== LEAK SUMMARY:
==630==   definitely lost: 400 bytes in 1 blocks
==630==   indirectly lost: 0 bytes in 0 blocks
==630==   possibly lost: 0 bytes in 0 blocks
==630==   still reachable: 0 bytes in 0 blocks
==630==   suppressed: 0 bytes in 0 blocks
==630==
==630== For lists of detected and suppressed errors, rerun with: -s
==630== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
rio_regex@DESKTOP-NVPM5HQ:~$
```

2.10.6 Performance Analysis

- CPU profiling dengan perf

```
rio_regex@DESKTOP-NVPM5HQ:~$ sudo apt install linux-tools-common linux-tools-generic
[sudo] password for rio_regex:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
linux-tools-common is already the newest version (6.8.0-85.85).
linux-tools-generic is already the newest version (6.8.0-85.85).
0 upgraded, 0 newly installed, 0 to remove and 28 not upgraded.
rio_regex@DESKTOP-NVPM5HQ:~$
```

- Monitor system calls

```
rio_regex@DESKTOP-NVPM5HQ:~$ sudo perf trace ls
WARNING: perf not found for kernel 6.6.87.2-microsoft

You may need to install the following packages for this specific kernel:
  linux-tools-6.6.87.2-microsoft-standard-WSL2
  linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

You may also want to install one of the following packages to keep up to date:
  linux-tools-standard-WSL2
  linux-cloud-tools-standard-WSL2
rio_regex@DESKTOP-NVPM5HQ:~$
```

- Top-like interface

```
rio_regex@DESKTOP-NVPM5HQ:~$ sudo perf top
WARNING: perf not found for kernel 6.6.87.2-microsoft

You may need to install the following packages for this specific kernel:
  linux-tools-6.6.87.2-microsoft-standard-WSL2
  linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

You may also want to install one of the following packages to keep up to date:
  linux-tools-standard-WSL2
  linux-cloud-tools-standard-WSL2
rio_regex@DESKTOP-NVPM5HQ:~$
```

BAB III

PENUTUP

Berdasarkan seluruh rangkaian praktikum Sistem Operasi yang telah dilakukan, dapat disimpulkan bahwa Linux memberikan pemahaman menyeluruh mengenai cara kerja inti dari sistem operasi, mulai dari layanan dasar hingga manajemen kernel. Melalui berbagai praktik, mahasiswa mempelajari fungsi utama sistem operasi seperti pengelolaan proses, sistem berkas, layanan sistem, serta pemanggilan sistem (system calls) yang menjadi jembatan antara aplikasi dan kernel. Selain itu, praktik tentang systemd, linker & loader, serta struktur OS memperlihatkan bagaimana Linux bekerja secara modular dan berlapis untuk menjaga efisiensi serta stabilitas sistem. Mahasiswa juga memahami bagaimana program dapat bersifat spesifik terhadap sistem operasi tertentu karena perbedaan format biner dan API. Praktik kernel module, proses booting, hingga debugging menggunakan alat seperti strace, ltrace, gdb, dan valgrind melatih kemampuan analisis serta troubleshooting. Secara keseluruhan, praktikum ini memperkuat pemahaman konseptual dan praktis tentang arsitektur, manajemen sumber daya, serta mekanisme internal sistem operasi Linux yang menjadi fondasi penting dalam dunia komputasi dan pengembangan perangkat lunak.